

Summary

This video elucidates key concepts in Engineering Mathematics, focusing on **Random Variables (RV)**.

- A **Random Variable** is a **mathematical function** mapping sample points from a sample space to probabilities.
- **Classification:**
 - **Discrete RV:** Countable outcomes, uses **Probability Mass Function (PMF)**
 - **Continuous RV:** Uncountable outcomes, uses **Probability Density Function (PDF)** with integration
- **Cumulative Distribution Function (CDF):** Maps probability "up to a point," forming a **staircase** for discrete variables
- **Key Rules:**
 - Probabilities must lie between 0 and 1
 - Total probability across all events = 1

1. Random Variable as a Mathematical Function

1.1 Core Validation Rules

1. **Range Constraint:** Output probability $\in [0,1]$
2. **Valid Mapping:** Each sample point maps to a single output probability; multiple inputs can share the same output, but one input cannot produce multiple outputs

1.2 Functional Mechanism

Component	Description
Input (x)	Sample Points (e.g., coin toss, dice roll)
Process (f)	Random Variable function
Output (y)	Resulting probability

2. Discrete Random Variables and Arrival Patterns

- Outcomes are **countable** and represented as **Probability Mass Function (PMF)**

2.1 Two-Coin Toss Example

Number of Heads (x)	Probability $P(X=x)$
0	$1/4$
1	$2/4 = 1/2$
2	$1/4$
Sum	1

2.2 Three-Coin Toss Example

x (Heads)	$P(X=x)$
0	$1/8$
1	$3/8$
2	$3/8$
3	$1/8$
Total	1

2.3 Identifying Discrete Phenomena

Examples:

- Average vehicles arriving
- Number of dots on a die
- Number of customers entering a showroom
- Selecting a specific card or ball

3. Continuous Random Variables and Measurement

- Outcomes are **uncountable**, representing **measurements**
- **Integration** replaces summation to calculate probabilities across intervals

3.1 Common Continuous Phenomena

- Battery discharge time
- Height, volume, area
- Delivery systems and environmental measurements

4. Cumulative Distribution Function (CDF)

- Denoted $F_X(x)$, probability $X \leq x$
- Describes probability "up to point x "

4.1 Staircase (Ladder) Function for Discrete Variables

- **Monotonically Increasing:** never decreases
- **Bounds:** $F_X(-\infty)=0$, $F_X(\infty)=1$
- **Positivity:** Always positive
- **Jump Discontinuity:** $P(X=x) = \text{Right Hand Limit} - \text{Left Hand Limit}$

4.2 CDF Example: Three Coin Toss

x	CDF $F(x)$
0	1/8
1	4/8
2	7/8
3	1

5. Comparative Summary Table

Feature	Discrete Random Variable	Continuous Random Variable
Nature of Data	Countable (Integers)	Uncountable (Real numbers / Decimals)
Key Function	PMF	PDF
Mathematical Tool	Summation (Σ)	Integration ($\int$)
Example	Number of heads, dice dots	Height, time, area, volume
Probability at a point	Exists as a specific value	Mathematically zero (requires interval)
CDF Shape	Staircase / Step function	Continuous curve

6. Future Directions

- Understanding Random Variables requires moving from **simple probability** to **functional analysis**
- **Discrete RVs**: Focus on integers and arrival patterns
- **Continuous RVs**: Focus on measurements and integration
- Progression leads to specialized distributions:
 - **Uniform RVs**: Equal probability outcomes (e.g., dice)
 - **Normal (Gaussian) RVs**: Limit of discrete trials as number of trials $\rightarrow \infty$
 - **Bivariate RVs**: Two-dimensional analysis for advanced applications